

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457397.915898 +/- 2.3e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.031
Transit depth (Rp/Rs)^2	1.73 +/- 0.83 [%]
Orbital Inclination (inc)	82.29 +/- 2.38
Airmass coefficient 1 (a1)	0.972 +/- 0.016
Airmass coefficient 2 (a2)	0.025 +/- 0.032
Scatter in the residuals of the lightcurve fit is	1.62 %
Best Comparison Star	#2 - [507, 85]
Optimal Aperture	3.34
Optimal Annulus	9.34
Transit Duration (day)	0.053 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.031 vs 0.1214 (deviation 0.28σ)
Duration fit vs published	0.053 d vs 0.0742 d (deviation 0.65σ)
Mid-time O-C	-0.3 min
Depth SNR	3.9
Residual scatter	1.62 %

Light curve

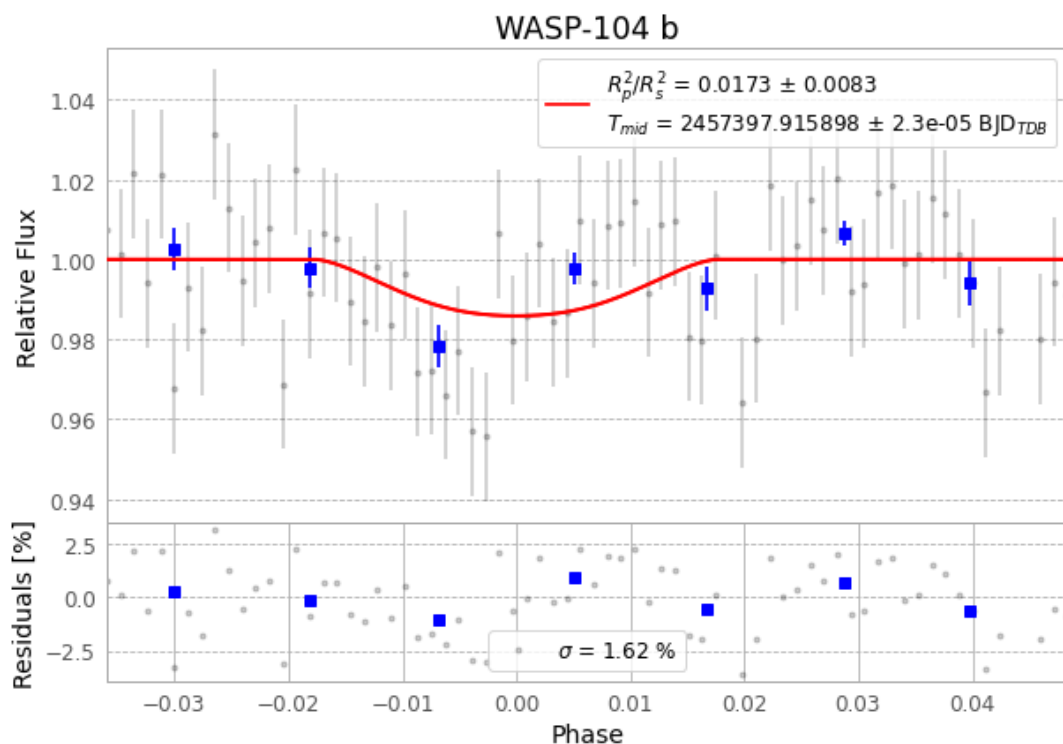


Figure 1 — FinalLightCurve_WASP-104 b_2016-01-10.png (SHA-256 a45d4608c181086b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a23f667eb097b693...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), WASP-104160110082112.FITS → WASP-104160110115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-160110.log (5e0d30d6a8dc...), FinalLightCurve_WASP-104 b_2016-01-10.png (a45d4608c181...), FinalLightCurve_WASP-104 b_2016-01-10.csv (b110fc205ddf...)

Compute resources

Wall time 102.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 01:15Z.